// MATHEMATICS LAB · GNU OCTAVE

Mapping the Invisible

Wi-Fi Signal Mapping and Router Placement

Vector calculus applied to wireless signal strength

$$S(x,y) = 100 \cdot e^{-((x-2)^2+(y-2)^2)/10} + 80 \cdot e^{-((x-8)^2+(y-7)^2)/10}$$

100.18	0.17	61 m²	19.27
peak at (2,2)	dead zone (0,10)	usable, S > 30	path average

LAB EXPERIMENTS USED

- Exp 2 · Surface & contour
- Exp 3 · Maxima / minima
- Exp 5 · Double integral
- Exp 7 · Gradient arrows
- Exp 8 · Line integral

SUBMITTED BY

Syed Abdullah Kazmi

B.Tech CSE (Cyber Security) · VIT Vellore

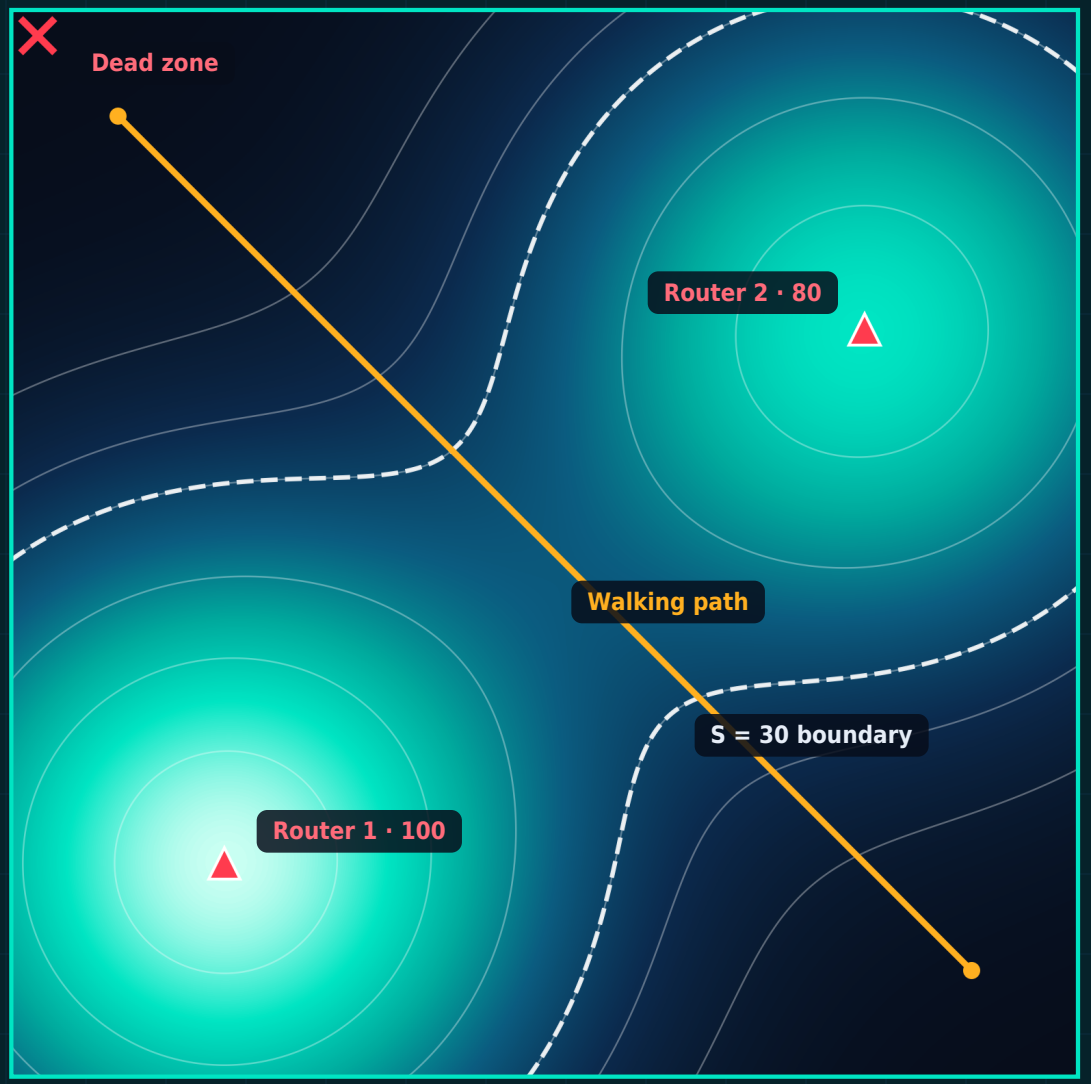


Fig. · Signal landscape S(x, y), 10 m × 10 m room

INSIDE: Signal model · Octave code · Output · Conclusion